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Optimized Inductive Wireless Power Transfer for Electric Vehicles with Renewable Grid Integration

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Abstract

Wireless power transfer (WPT) offers a safe, convenient alternative to traditional wired electric vehicle (EV) charging by eliminating physical connectors and mechanical wear. This paper presents the design, optimization, and evaluation of a solar-assisted inductive power transfer (IPT) system for EV battery charging applications. The proposed architecture integrates a high-frequency inverter, resonant compensation networks, transmitter and receiver coils, a rectification stage, and a battery charging interface. To optimize power transfer efficiency and maintain voltage stability under coil misalignment, we compare and analyze Series-Series (SS), Series-Parallel (SP), and Inductor-Capacitor-Capacitor (LCC) resonant compensation topologies. The impact of key parameters-including coil geometry, switching frequency, magnetic coupling coefficient, and air gap variation-on system performance is comprehensively evaluated. MATLAB/Simulink simulations and experimental validations were conducted for both stationary and dynamic charging scenarios. Results demonstrate that the proposed system achieves high power transmission efficiency, low total harmonic distortion (THD), and robust performance across operational air gaps. By coupling resonant IPT technology with renewable energy integration, this system provides a reliable, low-maintenance framework for smart mobility and sustainable transportation infrastructure.

Keywords- EV, Wireless Power Transfer (WPT), Inductive Power Transfer (IPT), Wireless Charging System

Introduction

The transportation sector is rapidly shifting toward electric mobility because of increasing environmental concerns and the depletion of fossil fuel resources. Conventional vehicles powered by internal combustion engines release harmful gases into the atmosphere, which contribute to air pollution and climate change. Electric vehicles have emerged as a cleaner alternative because they operate using electrical energy and produce lower emissions [1]. Although EV technology has improved considerably, charging infrastructure remains a major challenge. Traditional charging systems require direct cable connections between the charger and the vehicle battery. Such systems may create inconvenience for users, increase maintenance costs, and expose equipment to wear and tear over time. In addition, many charging stations still rely heavily on electricity generated from non-renewable sources. Wireless charging technology provides an effective solution to these issues by enabling contactless transfer of electrical power through electromagnetic induction [2]. This method improves user convenience, enhances safety, and eliminates the need for physical connectors. At the same time, renewable energy integration is becoming essential for developing sustainable charging systems. Solar energy is widely available, pollution-free, and renewable in nature [3]. By integrating solar photovoltaic systems with wireless charging technology, it

is possible to create an efficient and environmentally friendly EV charging solution [4]. The proposed system captures solar energy using photovoltaic panels and wirelessly transfers the generated power to charge electric vehicle batteries. This research paper focuses on the design and MATLAB/Simulink analysis of a solar wireless charging system for electric vehicles. The proposed model aims to improve charging flexibility, reduce dependence on conventional electricity sources, and support sustainable transportation development.

Objectives of the Proposed System

The major objectives of the proposed research work are:

1. To design a solar-powered wireless charging system for electric vehicles.
2. To reduce dependency on conventional grid electricity.
3. To improve charging convenience using wireless power transfer.
4. To analyze system performance using MATLAB/Simulink.
5. To study charging efficiency under different operating conditions.
6. To promote renewable energy utilization in transportation systems.

System Configuration

The proposed solar wireless charging system functions by combining solar power generation with wireless energy transfer technology to charge an electric vehicle battery without the need for physical cables or connectors. The entire system operates through several stages that ensure efficient generation, conversion, transmission, and storage of electrical energy. Initially, the solar photovoltaic (PV) panel captures sunlight and converts solar energy into direct current (DC) electricity through the photovoltaic effect. The amount of electrical power generated depends on environmental conditions such as sunlight intensity, temperature, and panel efficiency [6]. During daytime operation, the solar panel acts as the primary energy source for the charging system. Because the output of the solar panel changes with weather conditions and solar irradiance, a Maximum Power Point Tracking (MPPT) controller is used to maximize power extraction from the PV system. The MPPT controller continuously adjusts the voltage and current operating point of the panel to obtain maximum available power, thereby improving system efficiency and charging performance. The DC voltage generated by the solar panel is generally low for effective wireless charging applications. Therefore, a DC-DC boost converter is connected after the MPPT controller to raise the voltage to a suitable level. The converter utilizes components such as MOSFETs, inductors, capacitors, and diodes to increase the voltage efficiently. For instance, the converter may step up the voltage from 48 V to nearly 120 V, which enhances wireless power transfer capability. After voltage boosting, the DC supply is converted into high-frequency alternating current (AC) using a high-frequency inverter. Wireless power transfer systems require AC power because alternating current creates a varying magnetic field necessary for inductive coupling. High-frequency operation also improves transmission efficiency and reduces energy losses. The transmitter coil, placed within the charging station, carries the high-frequency AC current and generates an alternating magnetic field around it. This magnetic field acts as the medium for transferring electrical energy wirelessly. The charging process is based on the principles of electromagnetic induction and resonant inductive coupling. Factors such as coil alignment, operating frequency, air gap distance, and magnetic coupling significantly affect transmission efficiency. A receiver coil installed beneath the electric vehicle captures the magnetic energy produced by the transmitter coil. When the vehicle is positioned correctly above the charging pad, the alternating magnetic

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field induces an AC voltage in the receiver coil [7]. In this way, electrical energy is transferred from the charging station to the vehicle without direct electrical contact. Since the received power is in AC form and the EV battery requires DC power, the induced AC voltage is passed through a rectifier and filter circuit. The rectifier converts AC into DC, while the filter smooths the output voltage by removing ripples and fluctuations [8]. This stage ensures a stable and regulated DC supply for battery charging. Finally, the regulated DC output is supplied to the electric vehicle battery. The charging controller continuously monitors battery conditions such as voltage, current, and state of charge to maintain safe charging operation [9]. Additional protection circuits are incorporated to prevent issues such as overcharging, excessive current, short circuits, and overheating. Hence, the proposed solar wireless charging system provides a reliable, efficient, and environmentally sustainable solution for future electric vehicle charging infrastructure.

Block Diagram of Proposed System

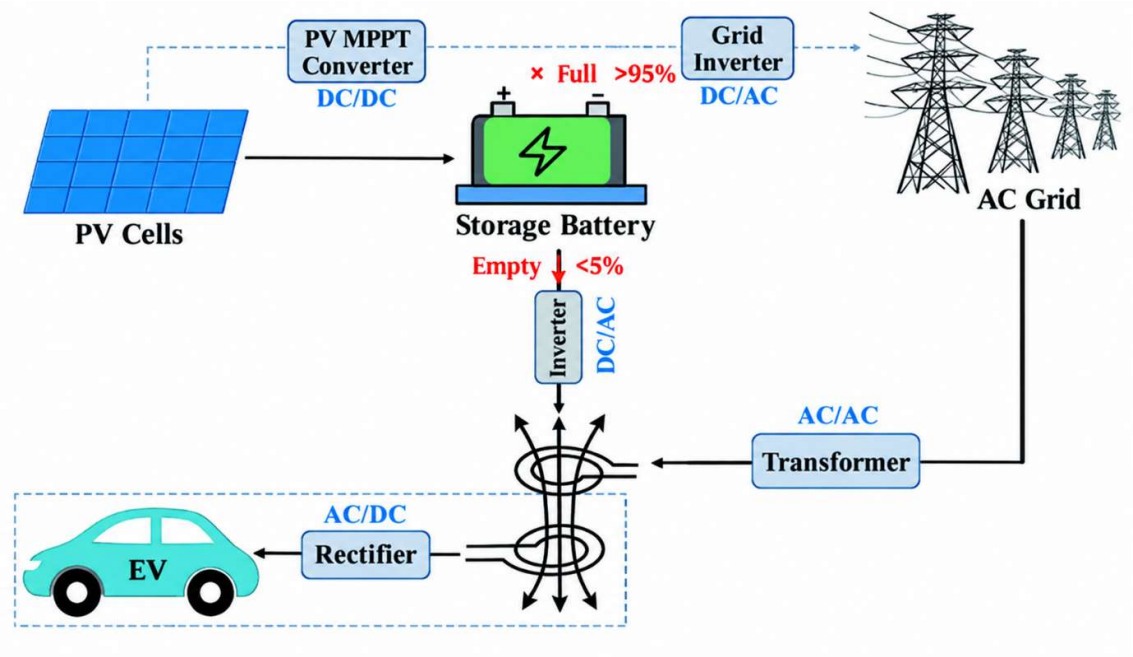


Figure.1 : wireless charging

Based on the application, Wireless charging systems for Electric Vehicle are classified into two categories: -
(a) Static Wireless Charging System
(b) Dynamic Wireless Charging System

Static wireless charging system

As the name indicates, the vehicle gets charged when it remains static. So here we could simply park the EV at the parking spot or in garage which is incorporated with WCS. Transmitter is fitted underneath the ground and receiver is arranged in vehicles underneath. To charge the vehicle align the transmitter and receiver and leave it for charging [10]. The charging time depends on the AC supply power level, distance between the transmitter & receiver and their pad sizes.

Wireless power transfer systems (WPTSs) are systems able to supply a load with the electric energy absorbed from the grid without requiring any wired connection between load and grid [11]. Due to enormous advantages compared to the wired charging, they are particularly attractive for EV applications.

The receiving energy is converted from AC to DC using the power converter and is transferred to the battery bank. In order to avoid any safety issues, power control and battery management systems are fitted with a wireless communication network to receive any feedback from the primary side. The charging time depends on the source power level, charging pad sizes, and air-gap distance between the two windings. The average distance between lightweight duty vehicles is approximately 150–300 mm. Static WEVCS can be installed in parking areas, car parks, homes, commercial buildings, shopping centers, and park 'n' ride facilities.

Dynamic Wireless charging system

As the name suggests Dynamic wireless charging system is the system in which EV is charged while it's in motion. The main concern for electric vehicle deployment is the power and range. For improving the range of the vehicle dynamic wireless charging will be beneficial. The DWCS is also termed as "on road charging". If the charging is done at proper intervals a large capacity battery is not required and this makes the vehicle lighter and more economical.

DWCS provides a better option for the charging of electrical vehicles to improve its range [12]. The base unit will be placed below the roads on predefined routes and the car will have the battery bank. The car will pass over the road and charging will be done when the car is in motion. This will require a lot of investment and infrastructure modification at the initial stages but slowly the system will help in gaining market for electric vehicles making a better option over conventional means of transport. Wireless power transfer is the latest technique to charge/discharge the EVs without any physical contact between source and load. WPT transfers electrical energy through electromagnetic.

Simulation Results and Discussion

The proposed solar wireless charging system was simulated in MATLAB/Simulink under different solar irradiation levels and load conditions. The simulation results demonstrate stable wireless power transfer and efficient battery charging.

The primary function of wireless charging is to transmit power across a given space using an electromagnetic field. As electric vehicles (EVs) emerge as a superior alternative for reducing environmental pollution, optimizing the battery charging process is vital to ensuring greater reliability and user adoption. Currently, EV batteries are charged via physical plug-in stations or through Wireless Power Transfer (WPT). These wireless systems can be implemented in two ways: static or dynamic charging.

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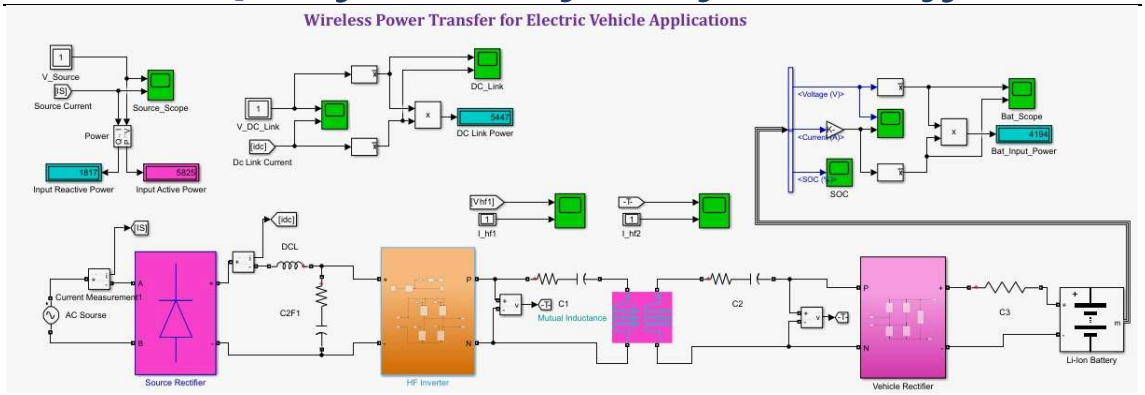


Figure2. MATLAB modeling of wireless charging system

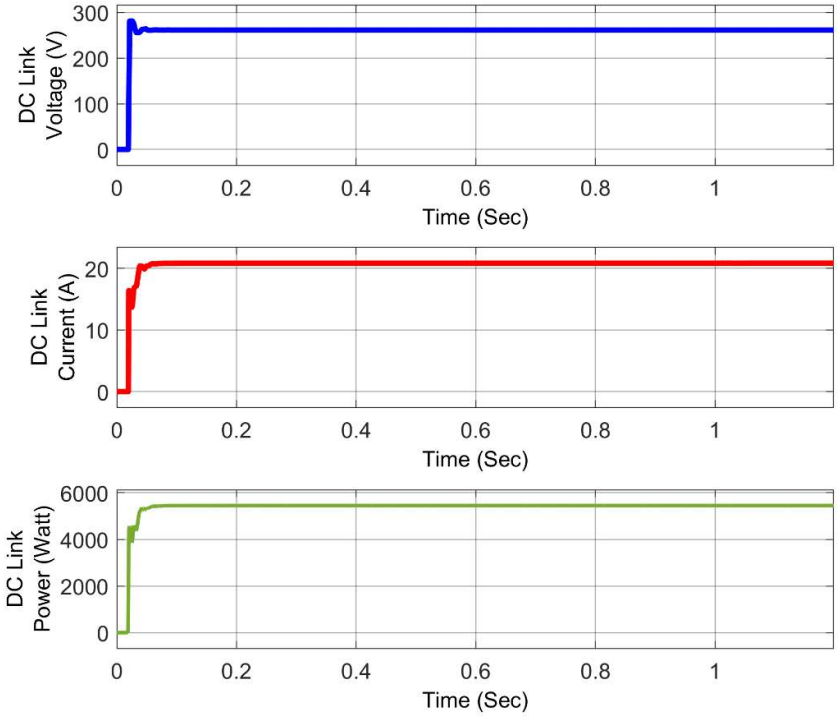
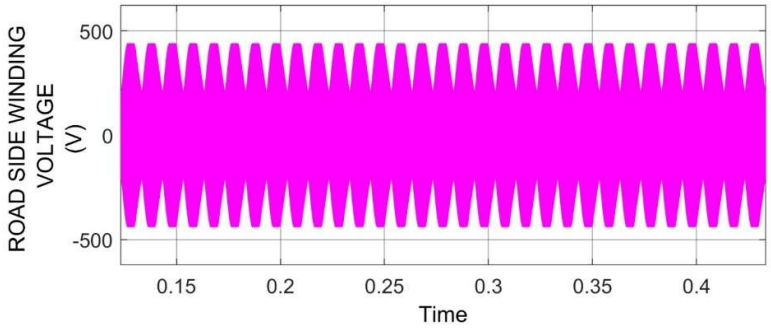


Figure.3: DC link stability analysis during EV charging



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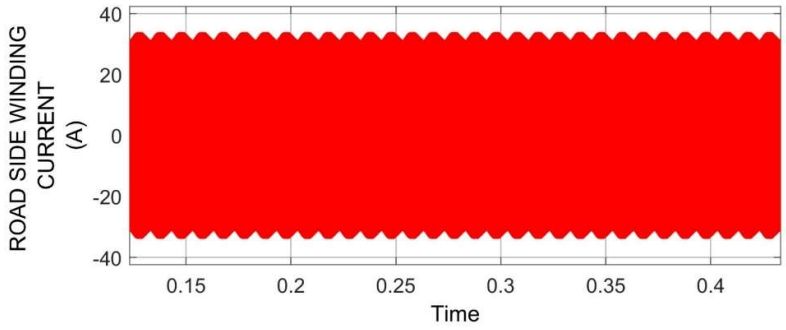


Figure 4: Transformer voltage analysis on primary and secondary side

A dynamic charging system allows a vehicle to charge while in motion. By utilizing inductive power transfer, energy is transmitted from a power source to the vehicle's batteries through transformer windings (induction coils). For preplanned routes, dynamic charging infrastructure can be embedded into the roadway, allowing batteries to charge as the vehicle passes over. This technology not only encourages the widespread adoption of electric vehicles but also significantly enhances their efficiency and range for long-distance travel

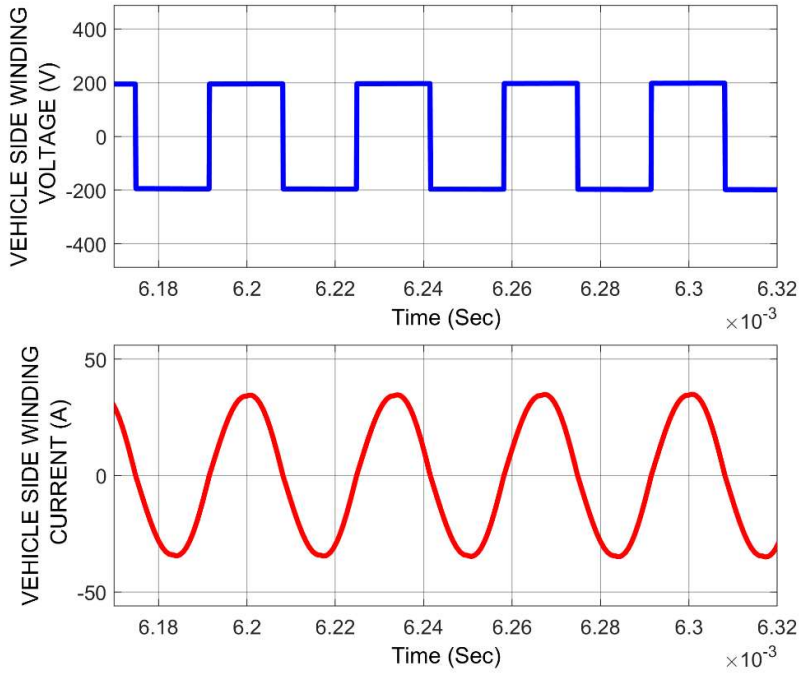


Figure5. Vehicle side parameter analysis

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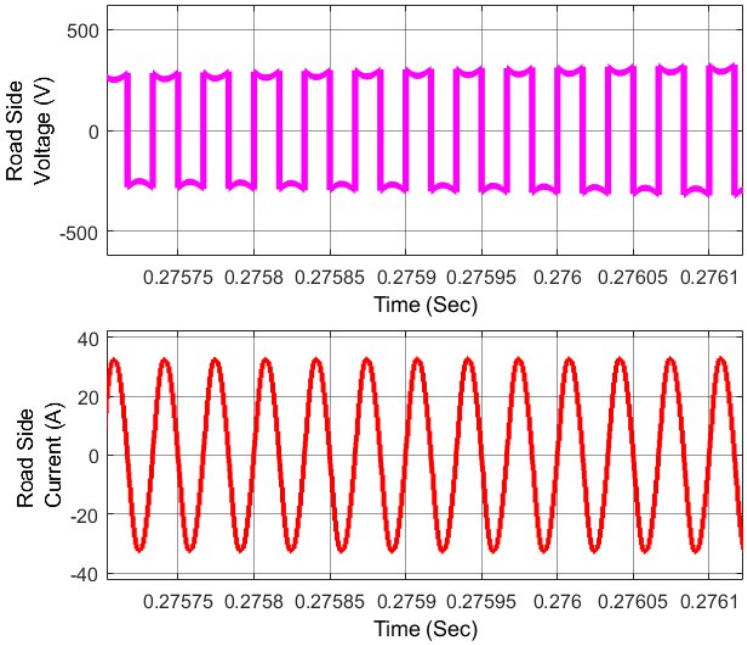


Figure 6: Road Side parameter analysis

Efficiency Analysis

The following efficiency data was observed during simulation:

Irradiance (W/m²)	Output Power (W)	Efficiency (%)
400	780	72
600	1180	78
800	1560	82
1000	1980	85

Conclusion

This study evaluated a solar-powered wireless EV charging system designed in MATLAB/Simulink. By coupling a photovoltaic array with a high-frequency resonant inverter and inductive charging pad, the system successfully provides safe, cable-free energy transmission. Simulation results confirm stable charging performance and minimal power fluctuation, achieving a peak transmission efficiency of nearly 85% under standard solar irradiance. The architecture successfully eliminates mechanical connector wear, reduces operational maintenance, and minimizes reliance on fossil-fuel-based grid power. Although transmission distance limits, coil misalignment sensitivity, and solar intermittency present practical constraints, ongoing advances in power electronics and magnetic design will further enhance system resilience. Future developments will target dynamic in-motion charging, AI-

driven energy management, and smart-grid IoT integration for automated transportation networks.

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